



Gennoor Tech
TRAIN. INNOVATE. BUILD.

5-DAY HANDS-ON BOOTCAMP · 40 HOURS

Agentic AI Engineering

Build production AI agents with LangChain, CrewAI, and MCP

5 40 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

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Course Overview

This 5-day hands-on bootcamp covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Python programming (intermediate), REST APIs, basic cloud familiarity.

Who Is This For?



Developers



AI Architects



ML Engineers



L&D Teams
Internal trainers

Learning Outcomes

- 1 **Build AI agents** — with LangChain, LangGraph, and tool calling
- 2 **Orchestrate multi-agent systems** — using CrewAI and Semantic Kernel
- 3 **Build MCP servers** — from scratch connecting AI to enterprise systems
- 4 **Deploy agents on** — Copilot Studio with Dataverse and Power Automate
- 5 **Design production agent** — architectures with memory, guardrails, and monitoring

5-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

AI Agent Foundations — LangChain & Tool Calling

HANDS-ON LABS + WORKSHOP

- SESSION A
- AI Agent Foundations
- Agent architectures: ReAct, Plan-and-Execute, Reflexion
 - LangChain core: chains, prompts, output parsers, memory
 - Building tool-calling agents with function calling

- SESSION B
- LangChain & Tool Calling
- Custom tools: web search, database queries, API calls
 - Lab: Build a research agent & SQL agent

02

LangGraph — Stateful Multi-Step Workflows

DEMOS + LABS + DISCUSSION

- SESSION A
- LangGraph
- Graph-based orchestration: nodes, edges, state management
 - Conditional routing, branching, and persistence (checkpoints)
 - Human-in-the-loop: approval gates, feedback loops

- SESSION B
- Stateful Multi-Step Workflows
- Error handling, retries, and streaming responses
 - Lab: Loan approval agent with human review step

03

CrewAI & Multi-Agent Collaboration

LECTURE + HANDS-ON LABS

SESSION A

CrewAI

- CrewAI architecture: agents, tasks, crews, processes
- Sequential vs hierarchical crew execution
- Semantic Kernel: plugins, planners, memory

SESSION B

Multi-Agent Collaboration

- AutoGen: multi-agent conversation patterns
- Lab: Financial research crew (analyst + writer + reviewer)

04

MCP Servers & Enterprise Integration

HANDS-ON LABS + WORKSHOP

SESSION A

MCP Servers

- Model Context Protocol: tools, resources, prompts
- Building MCP servers with Python and FastAPI
- Connecting MCP servers to SQL, Cosmos DB, and APIs

SESSION B

Enterprise Integration

- Copilot Studio integration with MCP and Power Automate
- Lab: Insurance claims processing agent (end-to-end)

05

RAG, Production Architecture & Capstone

DEMOS + LABS + DISCUSSION

SESSION A

RAG, Production Architecture

- RAG pipelines: chunking, embedding, indexing, hybrid search
- Azure AI Search, FAISS, Chroma integration
- Agent guardrails: content safety, prompt injection defense

SESSION B

Capstone

- CI/CD for agents: testing, versioning, deployment
- Capstone: Design and build a complete agentic AI system

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the bootcamp

Amazon Lex

AutoGen

Azure AI Search

Azure AI Services

ChromaDB

Cosmos DB

CrewAI

FAISS

FastAPI

LangChain

Microsoft Copilot

Microsoft Copilot Studio

Power Automate

Python

Semantic Kernel

What You Take Home

 **6 Agent Projects (complete source code)**
Included with course completion

 **Architecture Templates (BFSI, Government, Healthcare)**
Included with course completion

 **MCP Server Starter Kit (open-source)**
Included with course completion

 **Agent Design Canvas (roles, tools, memory, guardrails)**
Included with course completion

Certification of Completion

**Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "**Agentic AI Engineering**". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

AB-100
Agentic AI Business Solutions

AI-103
Azure AI Solutions

AI-901
Azure AI Fundamentals (New 2026)

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited
Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.
Builder of open-source MCP servers and Copilot Studio integrations. M.E. Mechanical Engineering, Mumbai University.

Delivery Options ■ **On-Site** — customized at your office | ■ **Virtual Live** — interactive via Teams/Zoom | 🌐 **International** — GCC, Africa, APAC, UK



Ready to Start Your Agentic AI Engineering Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

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